Individual Reflection

Introduction

This reflection evaluates my development of data analysis skills, statistical knowledge, and confidence using R throughout this module. Before starting the module, I had a theoretical understanding of descriptive and inferential statistics from my previous studies in Business Economics. However, I had limited practical experience applying statistical concepts through programming and analysing real-world datasets. Therefore, my main objective was to strengthen my analytical skills and understand how statistical methods can support evidence-based decision-making.

This reflection follows Rolfe et al.'s (2001) reflective framework by discussing what I experienced during the module, what I learned from these experiences, and how I will continue developing my skills.

Learning Activities and Experiences

Throughout this module, I developed practical experience using R for statistical data analysis. The activities included exploring datasets, preprocessing data, creating summary statistics, producing contingency tables, generating visualisations, performing hypothesis tests, correlation analysis, and regression modelling.

One of the most valuable aspects of the module was working with real-world datasets, including COVID-19, Health_Data, and diabetes datasets. These activities allowed me to apply statistical methods to meaningful topics and understand how quantitative analysis can contribute to research and decision-making.

I developed important skills in exploring and preparing datasets before performing analysis. Learning how to inspect variables, understand dataset structures, identify relevant information, and prepare data correctly was one of the most useful outcomes of the module because these steps are essential in any real-world analytical project.

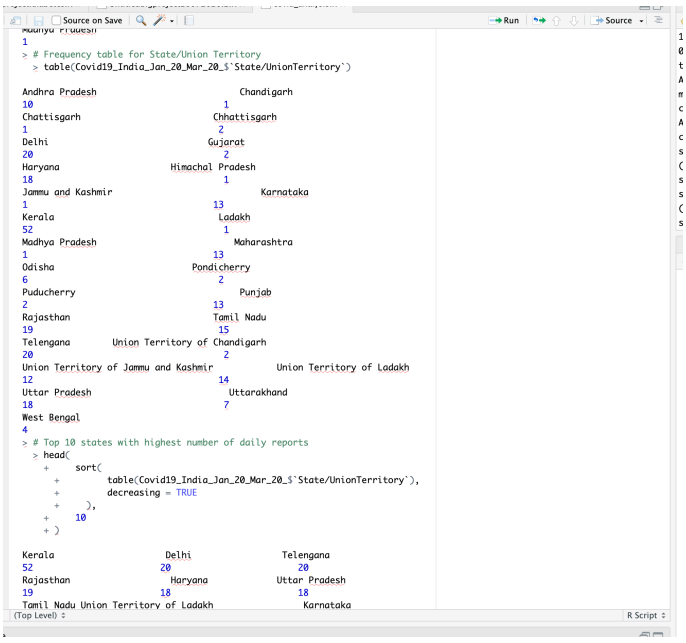


Figure 1 demonstrates an example of my initial dataset exploration using frequency tables in R. This activity helped me understand the importance of organising and examining data before applying more advanced statistical methods. I also gained experience producing summary tables and contingency tables, which improved my ability to compare variables, identify patterns, and communicate analytical findings clearly.

The practical exercises also helped me develop experience creating visualisations and interpreting statistical outputs. Working with healthcare-related datasets made statistical concepts more relevant and demonstrated how analysis can be used to investigate real problems.

Some datasets used during this module also became the foundation for additional experimentation with machine learning methods outside the module. This helped me recognise the relationship between statistics and artificial intelligence, where statistical knowledge provides the basis for reliable predictive modelling.

One challenge I experienced was related to dataset preparation and coding errors. Initially, I found preprocessing tasks and troubleshooting difficult because I had limited experience working with real datasets in R. To overcome these challenges, I used additional learning resources, followed R tutorials, reviewed documentation, and repeated practical exercises independently.

Reflection and Critical Analysis

Reflecting on my learning journey, I believe this module significantly improved my confidence using R. At the beginning, I relied heavily on examples, textbooks, and online tutorials to complete tasks. By the end of the module, I became more comfortable adapting code, solving errors independently, and applying statistical methods to different datasets.

Currently, I would evaluate my confidence using R at approximately 6 out of 10. I have developed a strong foundation in data exploration, preprocessing, statistical testing, and interpretation of results. However, I recognise that continued practice with larger and more complex datasets is required to become fully confident.

My previous studies provided me with knowledge of statistical concepts such as hypothesis testing, p-values, and confidence intervals. However, this module allowed me to transform theoretical knowledge into practical analytical skills.

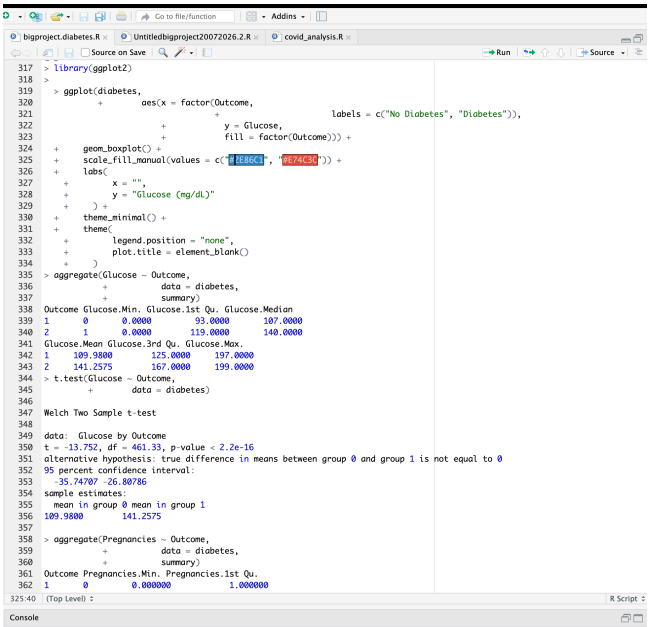
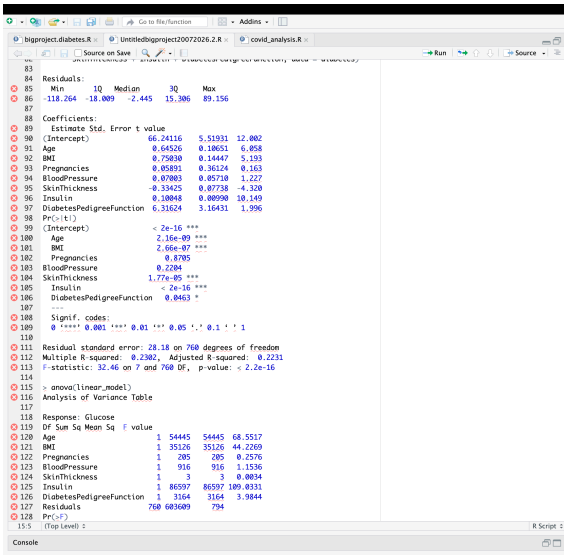


Figure 2 represents a Welch two-sample t-test comparing glucose levels between diabetes outcome groups. Working with this analysis improved my understanding that statistical interpretation involves more than identifying whether a p-value is significant. I learned to analyse confidence intervals, compare group differences, and consider whether findings have practical importance.

The weekly exercises, recommended readings, and mathematical activities reinforced the relationship between statistics and data science. I now understand that statistical reasoning is essential for responsible data analysis and accurate interpretation of analytical results.



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Residuals:
    Min      1Q  Median      3Q     Max
-118.264  -18.009   -2.445  15.306  89.156

Coefficients:
                          Estimate Std. Error t value
(Intercept)             66.24116    5.51931  12.002
Age                      0.64526    0.18651   6.858
BMI                      0.75030    0.14447   5.193
Pregnancies              0.05891    0.36124   0.163
BloodPressure            0.07003    0.05710   1.227
SkinThickness           -0.33425    0.07738  -4.320
Insulin                  0.10048    0.00990  10.149
DiabetesPedigreeFunction 6.31624    3.16431   1.996
Pr(>|t|)
(Intercept)              < 2e-16 ***
Age                      2.16e-09 ***
BMI                      2.66e-07 ***
Pregnancies              0.8705
BloodPressure            0.2204
SkinThickness           1.77e-05 ***
Insulin                  < 2e-16 ***
DiabetesPedigreeFunction 0.0463 *
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Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28.18 on 760 degrees of freedom
Multiple R-squared:  0.2302,  Adjusted R-squared:  0.2231
F-statistic: 32.46 on 7 and 760 DF,  p-value: < 2.2e-16

> anova(linear_model)
Analysis of Variance Table

Response: Glucose
                          Df Sum Sq Mean Sq  F value
Age                        1  54445   54445  68.5517
BMI                        1  35126   35126  44.2269
Pregnancies                1    205     205   0.2576
BloodPressure              1    916     916   1.1536
SkinThickness              1      3       3   0.0034
Insulin                    1  86597   86597 109.0331
DiabetesPedigreeFunction   1   3164    3164   3.9844
Residuals                760 603609     794
Pr(>F)
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Regression analysis was another important learning activity. As shown in Figure 3, I developed experience interpreting regression coefficients, p-values, R^2 values, and ANOVA results. These techniques improved my understanding of how multiple variables can influence an outcome and how statistical models can be used to investigate relationships within datasets.

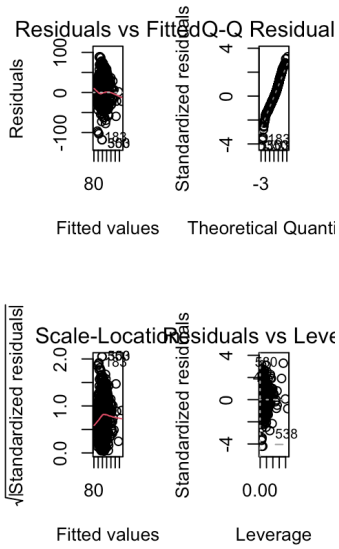


Figure 4 shows regression diagnostic plots used to evaluate model assumptions. Initially, interpreting these plots was challenging, but they improved my understanding of residual behaviour,

normality, homoscedasticity, and influential observations. I learned that creating a model is only one stage of analysis; evaluating whether the model is appropriate is equally important.

Feedback from weekly activities and tutor guidance helped me identify areas for improvement, particularly in explaining statistical results and developing greater confidence when interpreting outputs. This encouraged me to become more independent and critical in my approach to data analysis.

Another important reflection concerns my programming development. At the beginning of the module, I felt that my programming experience, particularly in Python, was limited. Instead of discouraging me, this motivated me to continue learning independently. I realised that developing technical skills requires persistence, problem-solving, and continuous practice.

Professional Development and Future Actions

This module provided me with a strong foundation in statistical programming and practical data analysis. My Professional Skills Development Plan highlights improvements in areas including critical thinking, problem-solving, digital skills, research, communication, and numerical analysis.

Skill	Evidence from the module	Future development
Time Management	Completed weekly practical activities alongside assessments	Continue planning study time more efficiently
Critical Thinking	Interpreted statistical findings using p-values, confidence intervals and regression outputs	Apply these skills to more complex datasets
Communication	Produced summary tables, plots and interpreted statistical outputs	Improve presentation and data storytelling skills
IT & Digital Skills	Used R for statistical analysis, preprocessing and modelling	Continue developing R and Python skills
Numeracy	Applied descriptive and inferential statistics	Study more advanced statistical methods
Research	Used academic literature and external tutorials to strengthen understanding	Continue independent research
Problem Solving	Debugged R code and resolved syntax errors independently	Improve coding efficiency
Ethical Awareness	Recognised the importance of accurate interpretation and responsible use of statistical findings	Apply ethical principles in future analytical work

The knowledge developed throughout this module will be valuable for my future career in artificial intelligence because machine learning models depend on effective data preparation, statistical reasoning, and accurate interpretation of results.

My next objective is to continue improving my R and Python skills through additional projects and independent learning. My long-term goal is to progress from a basic-intermediate level towards

advanced analytical capabilities that will allow me to work confidently with complex datasets in professional environments.

Although additional practical demonstrations during some sessions could have supported my learning, overcoming challenges independently strengthened my autonomy, resilience, and confidence as a learner.

Conclusion

Overall, this module significantly improved my understanding of statistical analysis and introduced me to the practical application of R as a statistical programming language. Although I initially faced challenges related to programming, dataset preparation, and troubleshooting, these experiences encouraged me to become a more autonomous learner.

I developed valuable skills in exploring datasets, preprocessing data, producing tables and visualisations, interpreting statistical findings, and evaluating regression models. Most importantly, I now have a clearer understanding of how statistics, programming, and data analysis work together to support reliable evidence-based decision-making. I intend to continue developing these skills and applying them to future data science and artificial intelligence projects.

References

Rolfe, G., Freshwater, D. and Jasper, M. (2001) *Critical Reflection in Nursing and the Helping Professions: A User's Guide*. Basingstoke: Palgrave Macmillan.

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Wickham, H. and Grolemund, G. (2017) *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Sebastopol, CA: O'Reilly Media.